

Towards a standard for exchanging passive acoustic monitoring (PAM) data

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Photo by Julia Wiel



RESEARCH INSTITUTE
NATURE AND FOREST





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- Open data
- Machine observation of biologging data
- Co-maintainer of Camtrap DP



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*Researcher data scientist
Instituto Humboldt*

- Artificial Intelligence
- Applied mathematics



Julia Wiel

*PhD student
NINA*

- Open science and standardisation
- Bioacoustics and PAM



Esteban Marentes

*Researcher data
administrator
Instituto Humboldt*

- Data quality
- Open science and standardisation



Juan S. Ulloa

*Researcher
Instituto Humboldt*

- Biodiversity Conservation
- Artificial Intelligence
- Acoustics

Overview

- Context 20 mins
- Hands-on workshop 80 mins
- How to contribute 20 mins





Colombian Ecoacoustic Network

We collect large volumes of data, but it's mostly shared through conferences and publications — not on GBIF.

Combining data from different projects remains challenging, making it hard to collaborate.

Our data remains fragmented and disconnected.



Project: Safe & sound

- Review of existing standards and guidelines for bioacoustics/PAM
- Reach out to users
- Collect use cases



WILDLABS.NET
[The conservation technology network]

arm

Conclusions

- Need for a widely accepted bioacoustics data standard that is:
 - widely accepted
 - not too complex
 - for both terrestrial and marine habitats
- Camtrap DP fits and can be extended for acoustic data



Camera Trap Data Package



Camtrap DP

Data exchange format for camera trap data

- <https://camtrap-dp.tdwg.org/>
- Bubnicki et al. 2023 [10.1002/rse2.374](https://doi.org/10.1002/rse2.374)¹
- Community-developed
- Maintained

FAIR

- Findability
- Accessibility
- Interoperability
- Reusability

¹Remote Sens Ecol Conserv, Volume: 10, Issue: 3, Pages: 283-295, First published: 09 December 2023

Camtrap DP software

- [camtrapdp](#) (R): read, edit and write
- [camtraptor](#) (R): explore and visualize
- [camtrapDensity](#) (R): REM and density calculation
- [Biowatch](#) (desktop prototype): explore and visualize
- [Integrated Publishing Toolkit \(IPT\)](#) (online): publish datasets to GBIF
- [Agouti](#) (online): data management
- [TRAPPER](#) (online): data management
- [EuroCaM](#) (online, *pending*): data management
- [Wildlife Insights](#) (online, *pending*): data management
- [WildObs](#) (online, *pending*): data management



Camtrap DP is a GBIF publication format

GBIF has adopted Camtrap DP

- automatic data transformation to Darwin Core and EML
- download dataset in original (richer) Camtrap DP

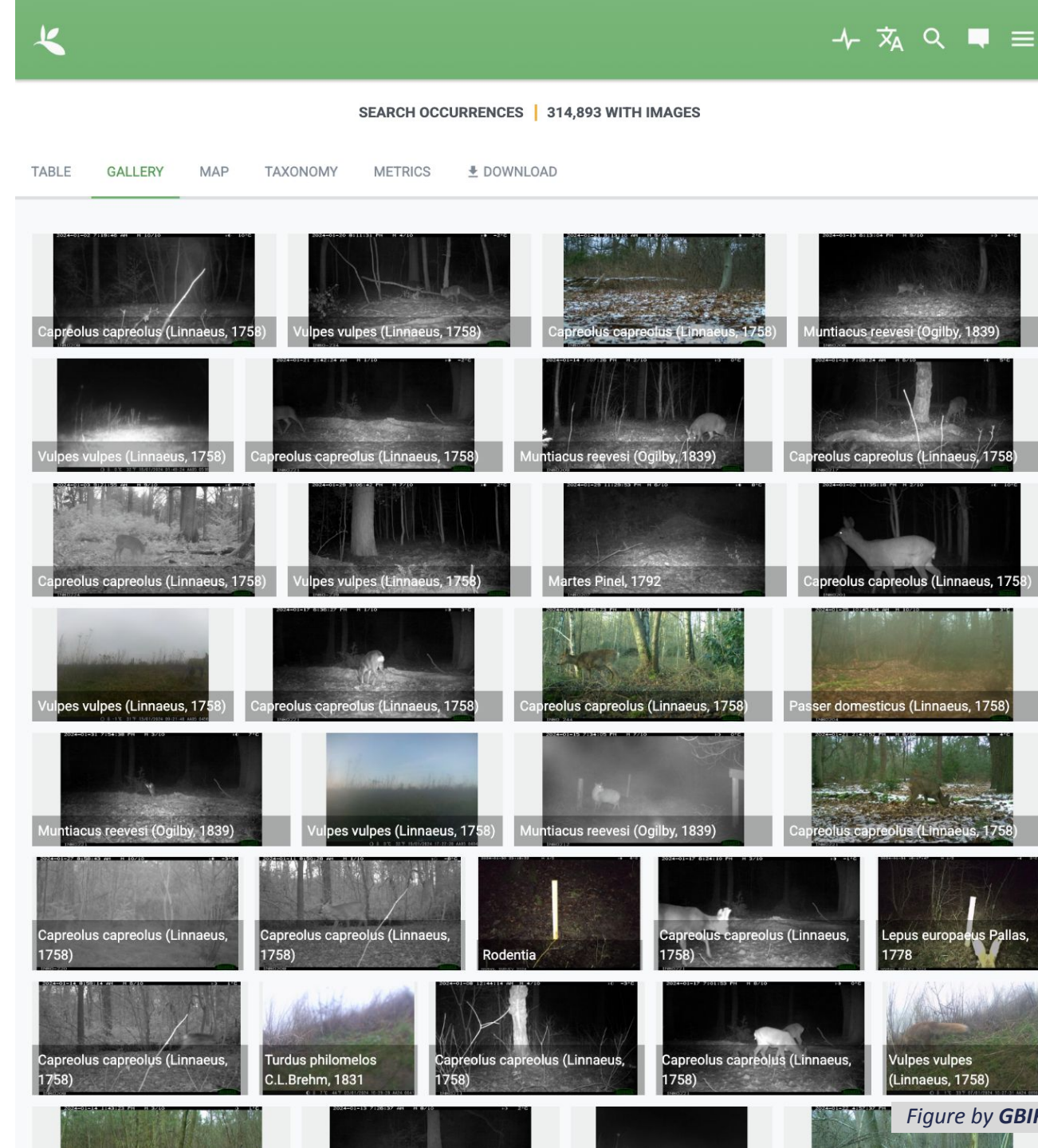


Figure by GBIF

https://www.gbif.org/occurrence/gallery?protocol=CAMTRAP_DP

Structure of Camtrap DP

Frictionless Data Package¹

4 files:

- datapackage.json
- deployments.csv
- media.csv
- observations.csv

¹ <https://datapackage.org/>



Pristimantis bogotensis, photo by Nicolás Betancourt

A

Project / Study

Deployment
29b7d356
from T1 to T1000



Media file
59b38bc6 at T200



Media file
cc50edaa at T201

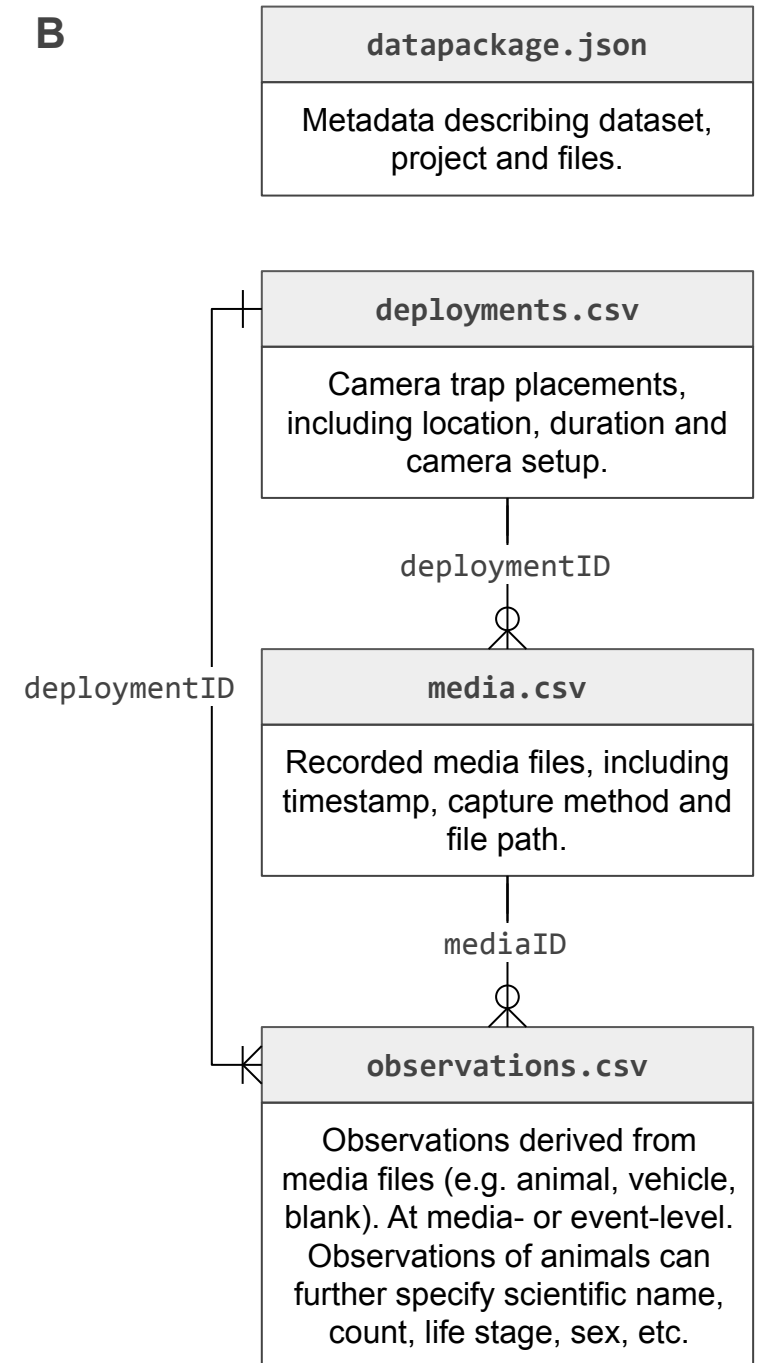


Media file
0966e552 at T202



observation ID	observation Level	media ID	event Start	event End	scientificName	count
obs1	media	59b38bc6	T200	T200	Anas platyrhynchos	3
obs2	media	cc50edaa	T201	T201	Anas platyrhynchos	4
obs3	media	0966e552	T202	T202	Anas platyrhynchos	4
obs4	event		T200	T202	Anas platyrhynchos	5

B





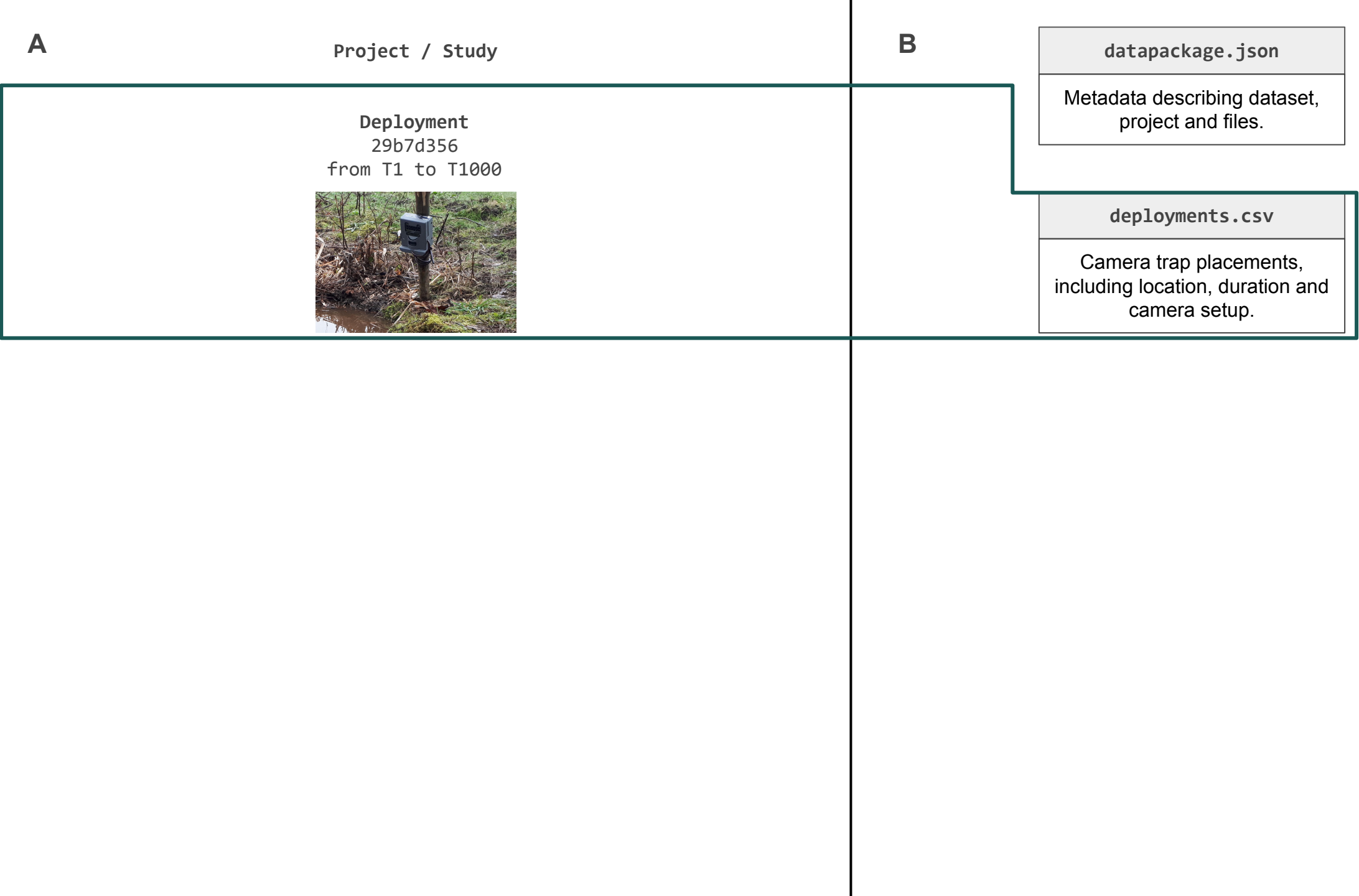


Figure by
Bubnicki et al. 2023

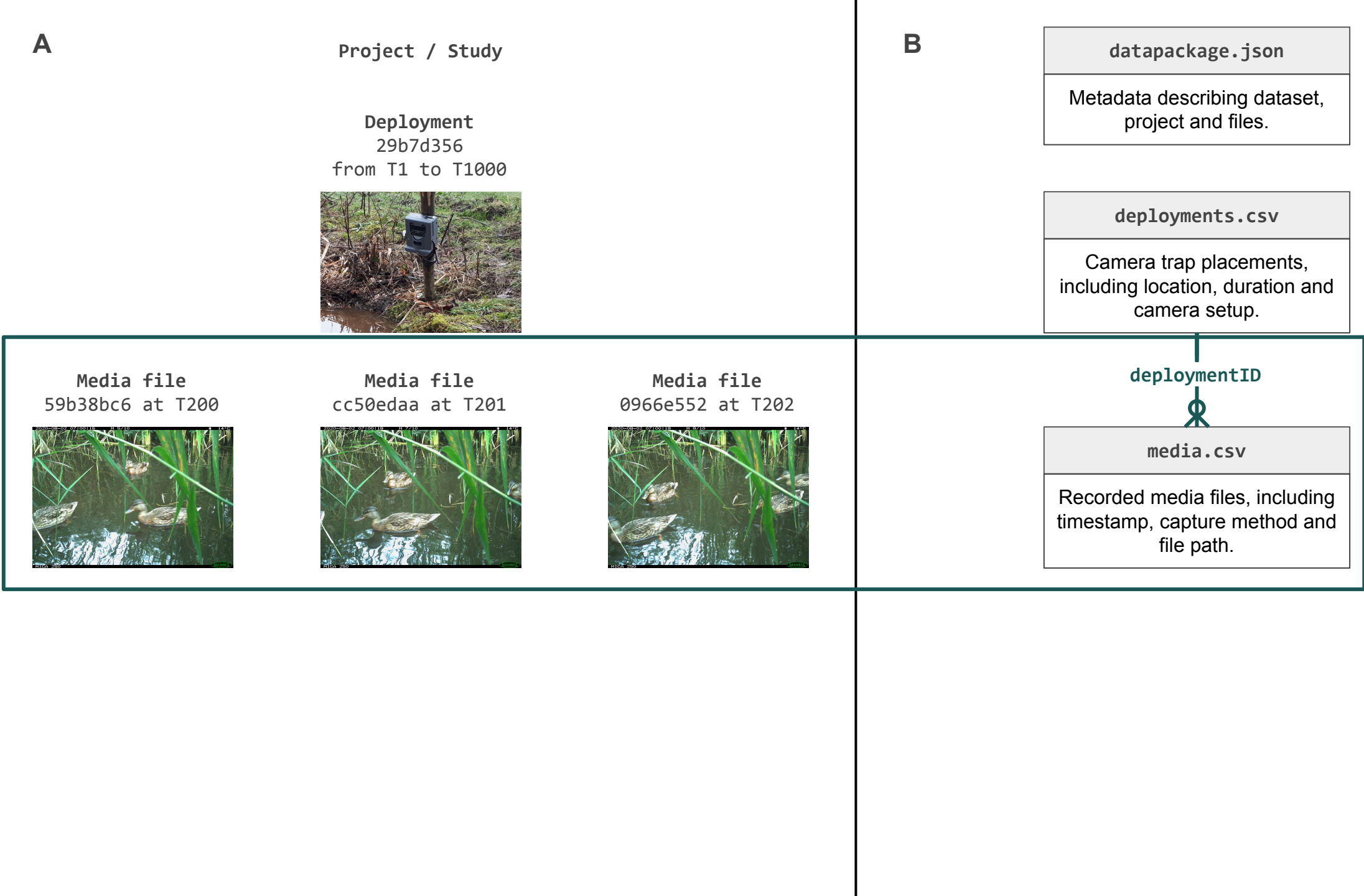


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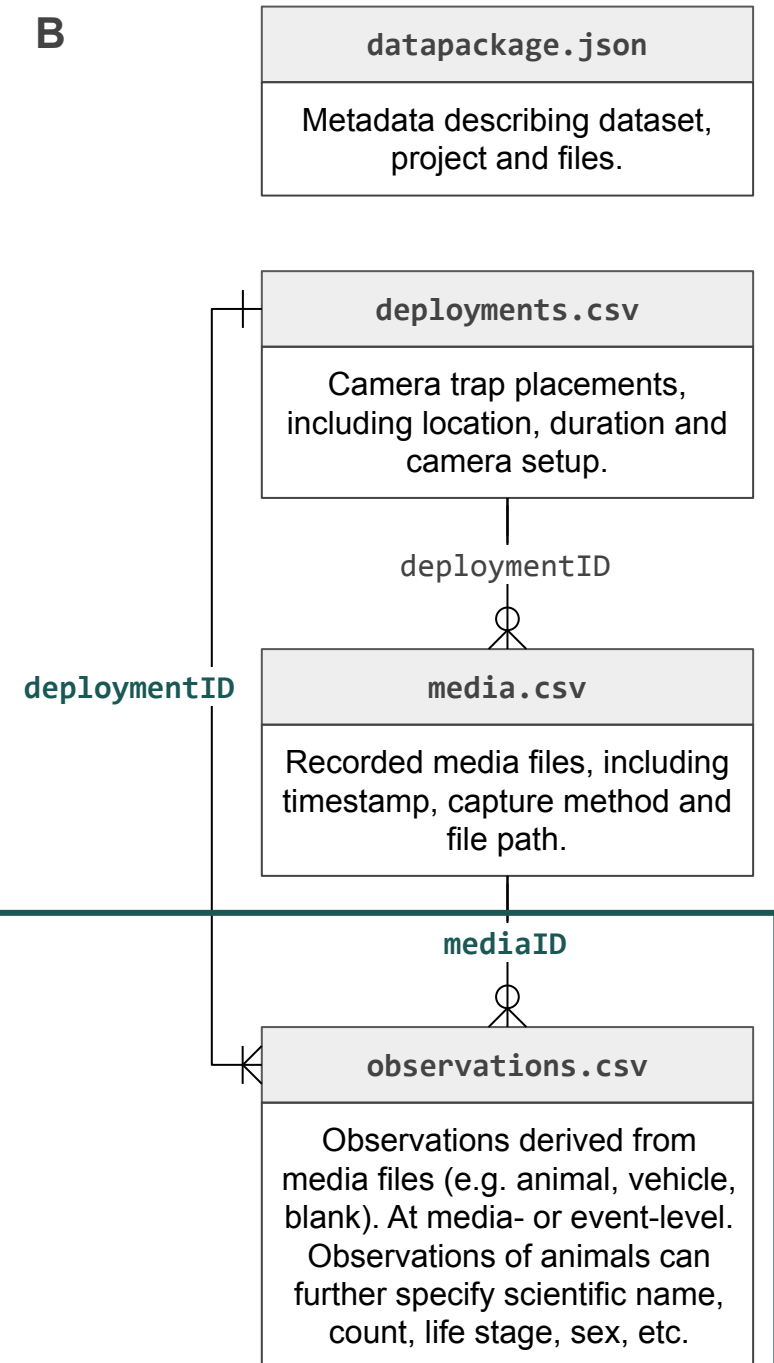


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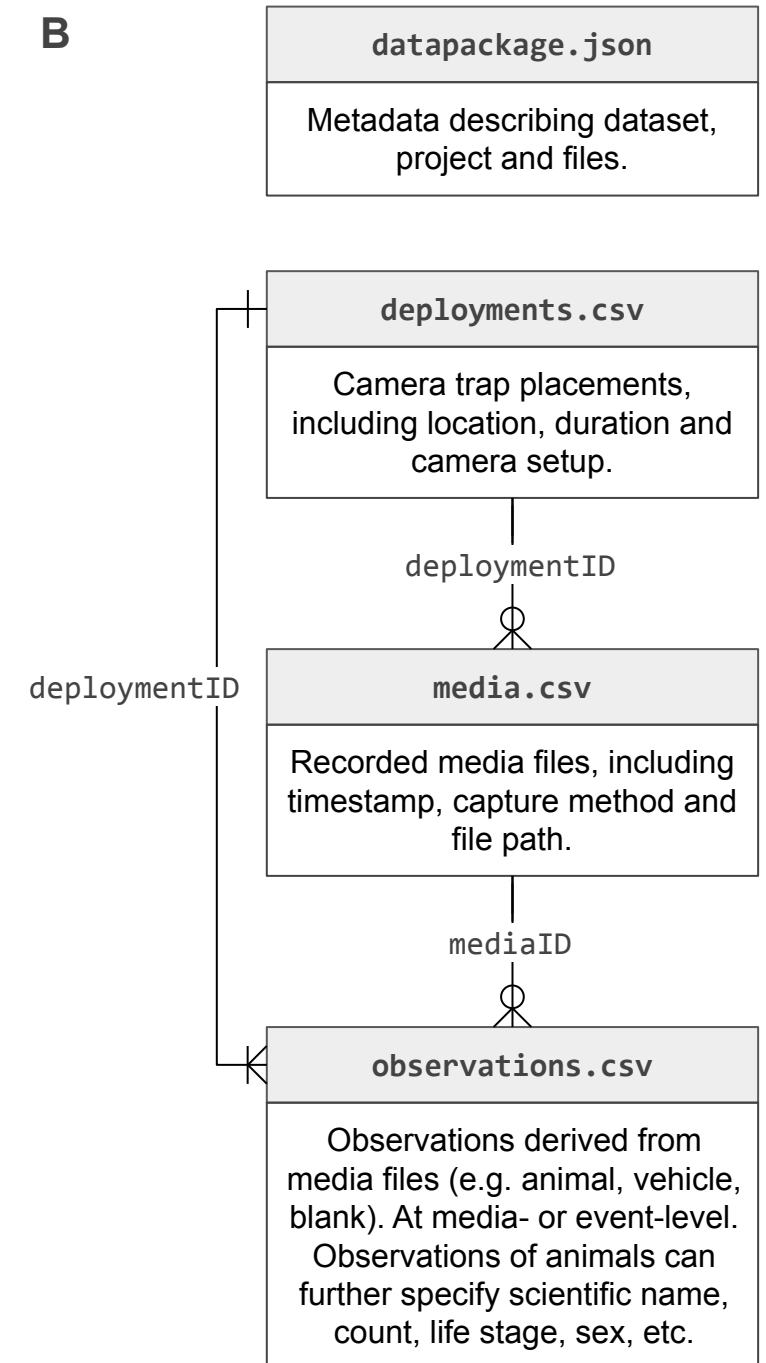


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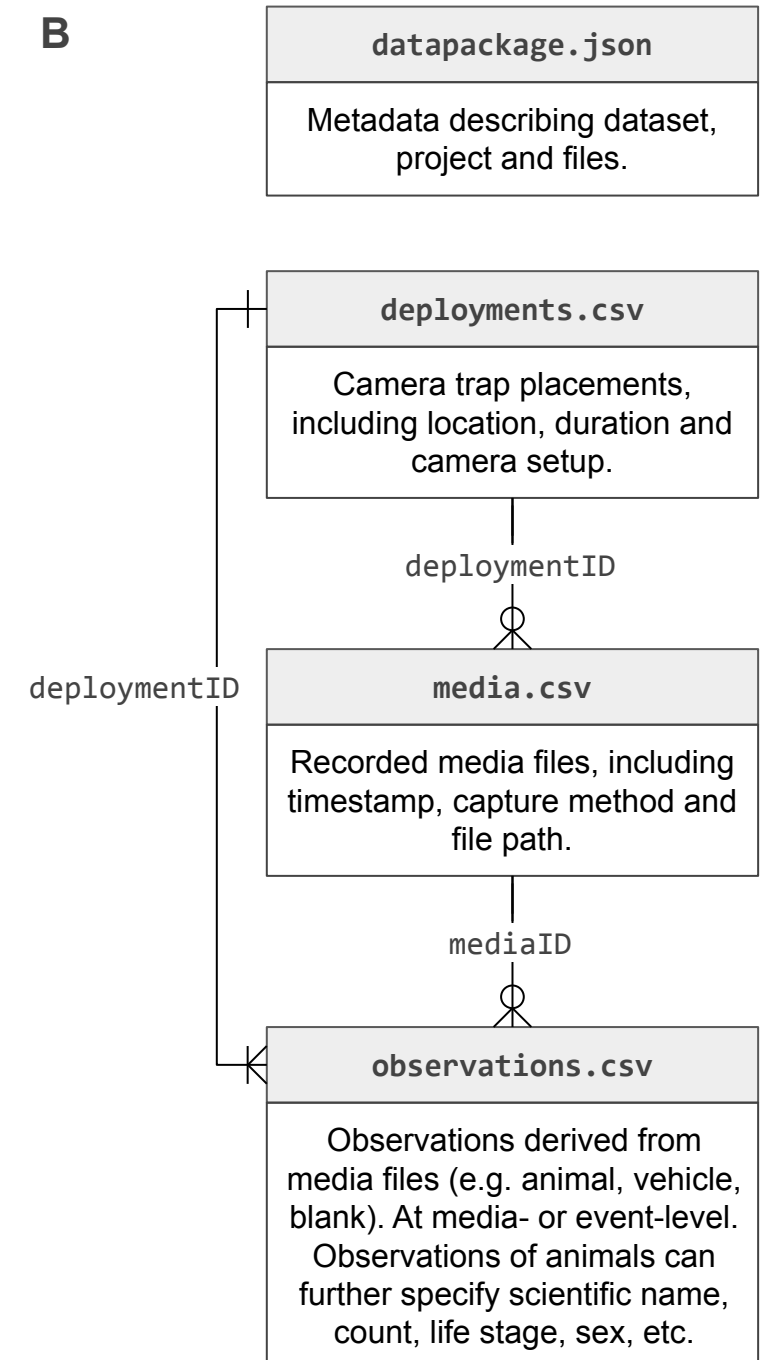


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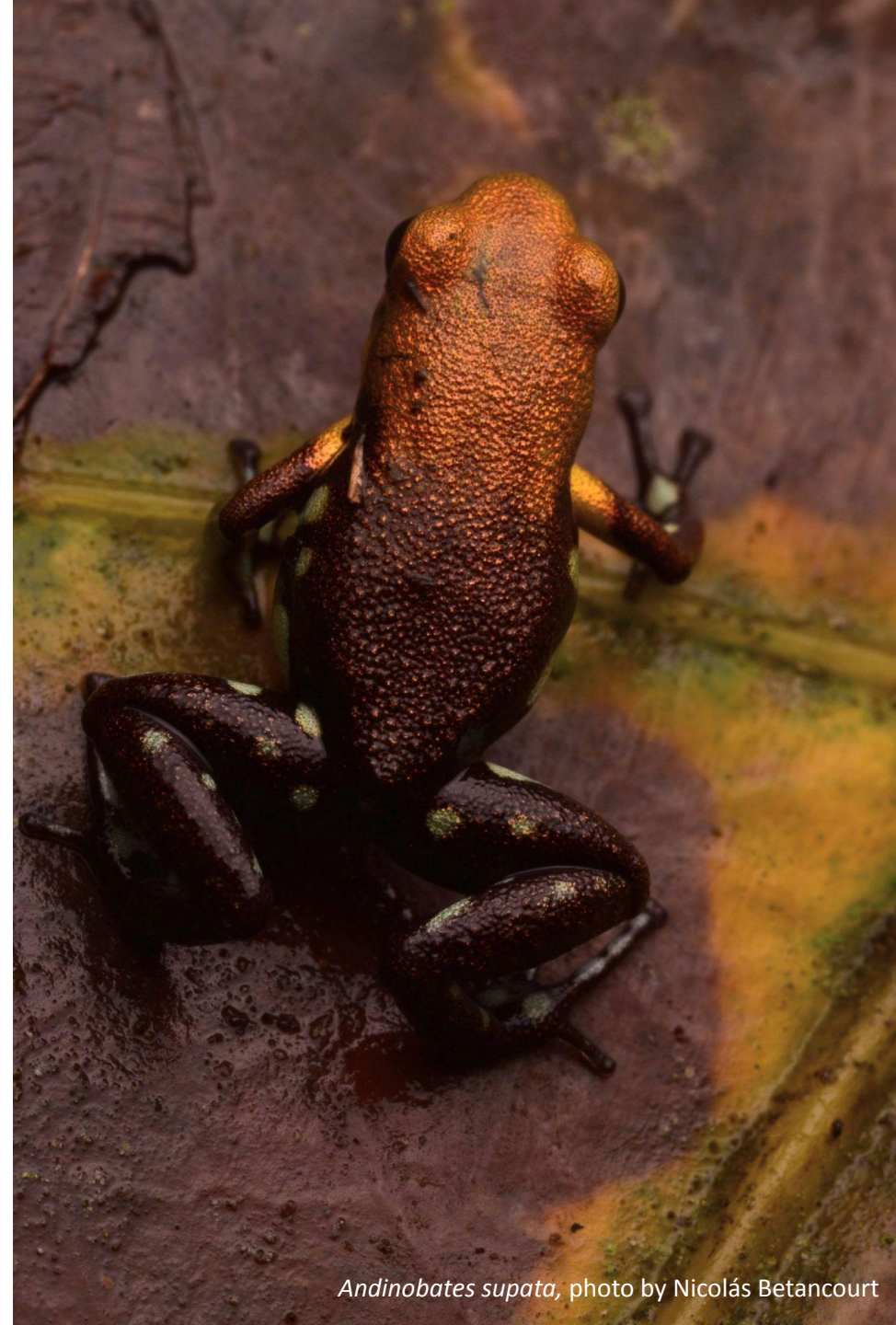
Camtrap DP is extensible

A Camera Trap Data Package
"camtrap-dp-example-dataset" with 3 tables:

- **deployments**: 4 rows
 - **media**: 423 rows
 - **observations**: 549 rows
- } **standardized**

And 1 **additional resource**:

- individuals
- } **not standardized**



Andinobates supata, photo by Nicolás Betancourt

Camtrap DP goes bioacoustics

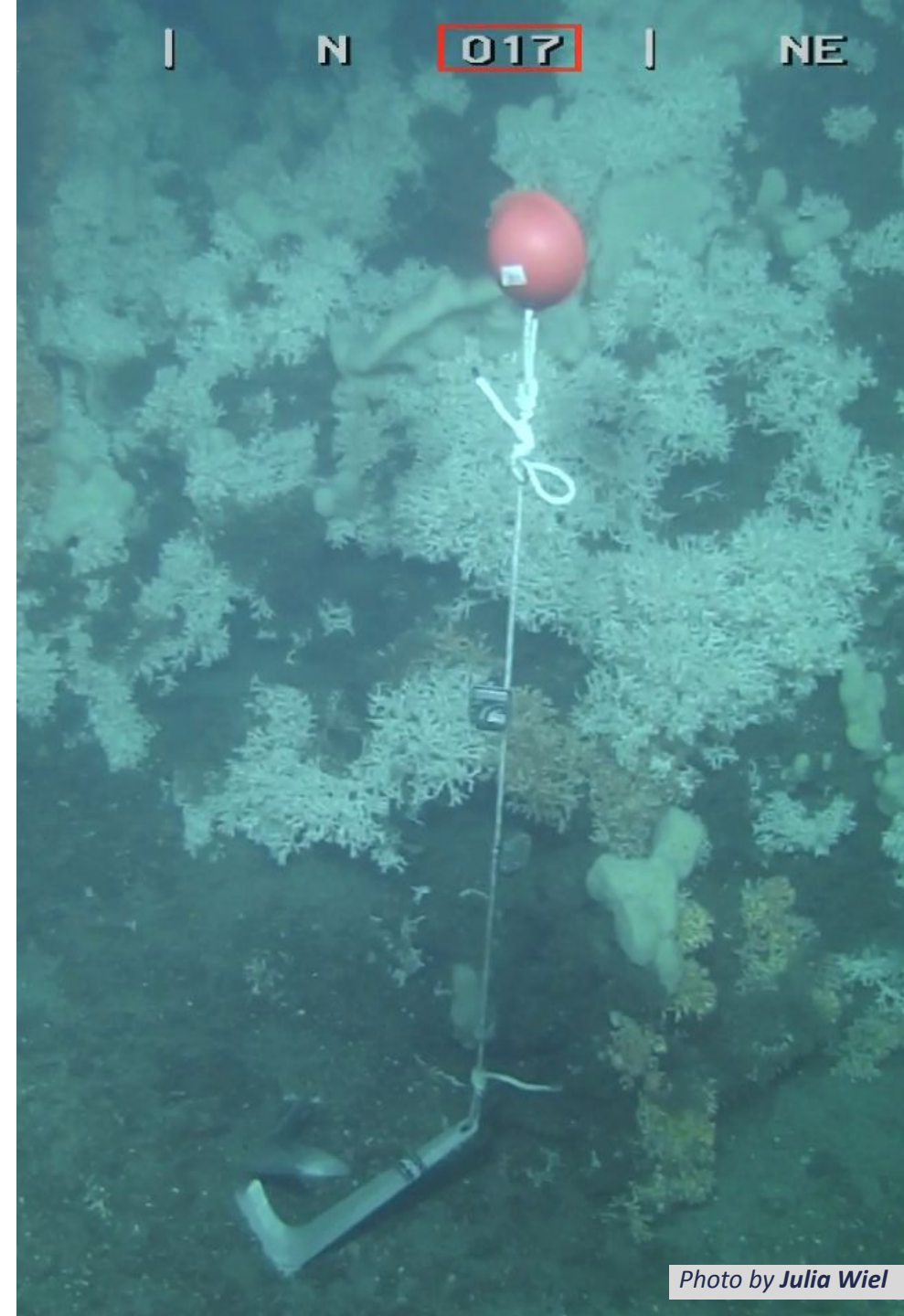
1) Change names

- cameraModel → deviceModel

2) Add new terms

- samplingFrequency
- gain
- channels

3) Extend descriptions



Next steps

- Finish [summary report](#)
- Continue the discussion in [GitHub issues](#)¹
- **Testing, testing, testing!**

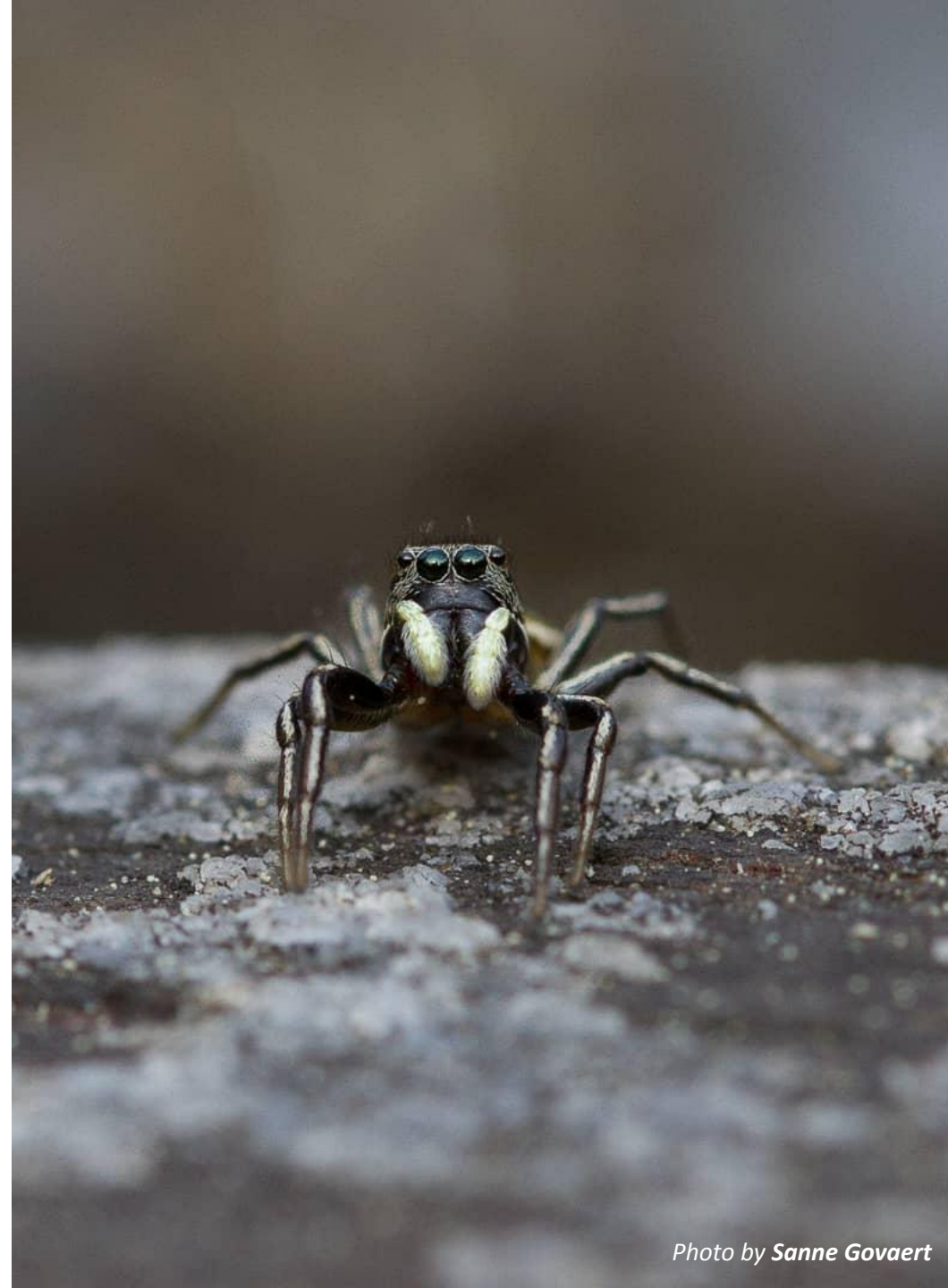
Mapping datasets to the suggested version of Camtrap DP

- Prepare the new version of Camtrap DP

Release of a new version of the standard is dependent on:

- How fast we get to consensus
- Funding

¹ <https://github.com/tdwg/camtrap-dp/issues>



is:issue state:open

Labels

Milestones

New issue

☐	Open 39	Closed 242	Author	Labels	Projects	Milestones	Assignees	Types	Newest
☐		Other formats versions of bounding boxes and such	#416 · quitmeyer opened 2 weeks ago						
☐		How are events defined faq	Task #415 · peterdesmet opened last month					2	
☐		Files at raw.githubusercontent.com return 404 bug	#414 · peterdesmet opened on Aug 8					1	
☐		Release steps documentation	#409 · peterdesmet opened on Jun 10					1	
☐		Add "interacts with camera" field community:feedback level:observations term:new	Feature #406 · Tim-Hofmeester opened on Apr 30					2	
☐		LinkML Model	#403 · dnlbauer opened on Apr 15					1	
☐		Controlled vocabulary for the "attractantType" field in the deployments table	#402 · kkuczkowska opened on Mar 28						

Hands-on Workshop

Workshop

Primary goal

> Understand and apply Camera Trap Data Package with the suggested changes

Mapping exercise:

- Map demo data to the 3 core tables (deployments, media, observations)
- Metadata files are excluded from this mapping exercise

Important

Our aim is that this bioacoustic data standard will be adopted in data management systems and databases, as it it also happens for camera trap data.

Demo dataset: The Sound of Norway

- autonomous acoustic monitoring network
- Jul–Nov 2021
- real-time transmission from field to google cloud (via mobile network)
- audio processed using birdnet

Adapted dataset for demo:

<https://github.com/camera-traps/bioacoustics/tree/main/datasets/living-data-workshop/data/raw>



30 sites

135,569 hrs

1,062,471
detections

56 species

Discover More

How?

2 groups:

- 1) no-coding group
- 2) coding with R
 - Make use of an existing .Rmd script to create a valid Camera Trap Data Package

Links for workshop

Summary report: <https://tinyurl.com/bioacoustic-camtrapdp>

Data:

<https://github.com/camera-traps/bioacoustics/tree/main/datasets/living-data-workshop/data/raw>

R:

- Download the whole repo and open the R project:
<https://github.com/camera-traps/bioacoustics>
- The script is at
[datasets/living-data-workshop/src/acoustic_to_camtrapdp.rmd](https://github.com/camera-traps/bioacoustics/tree/main/datasets/living-data-workshop/src/acoustic_to_camtrapdp.rmd)

Demo dataset

- Data was collected with Bugg v2022, attached to a pole at 1.5 m.
- Recording was continuous, and each media file was split in snippets of 5 min.
- The audio files are publicly available and contain some metadata. The files are 32 bit-float.

A solution

Sound of Norway in acoustic version of Camtrap DP

https://github.com/camera-traps/bioacoustics/blob/main/datasets/living-data-workshop/src/acoustic_to_camtrapdp_solution.rmd

<https://github.com/camera-traps/bioacoustics/tree/main/datasets/living-data-workshop/data/processed>

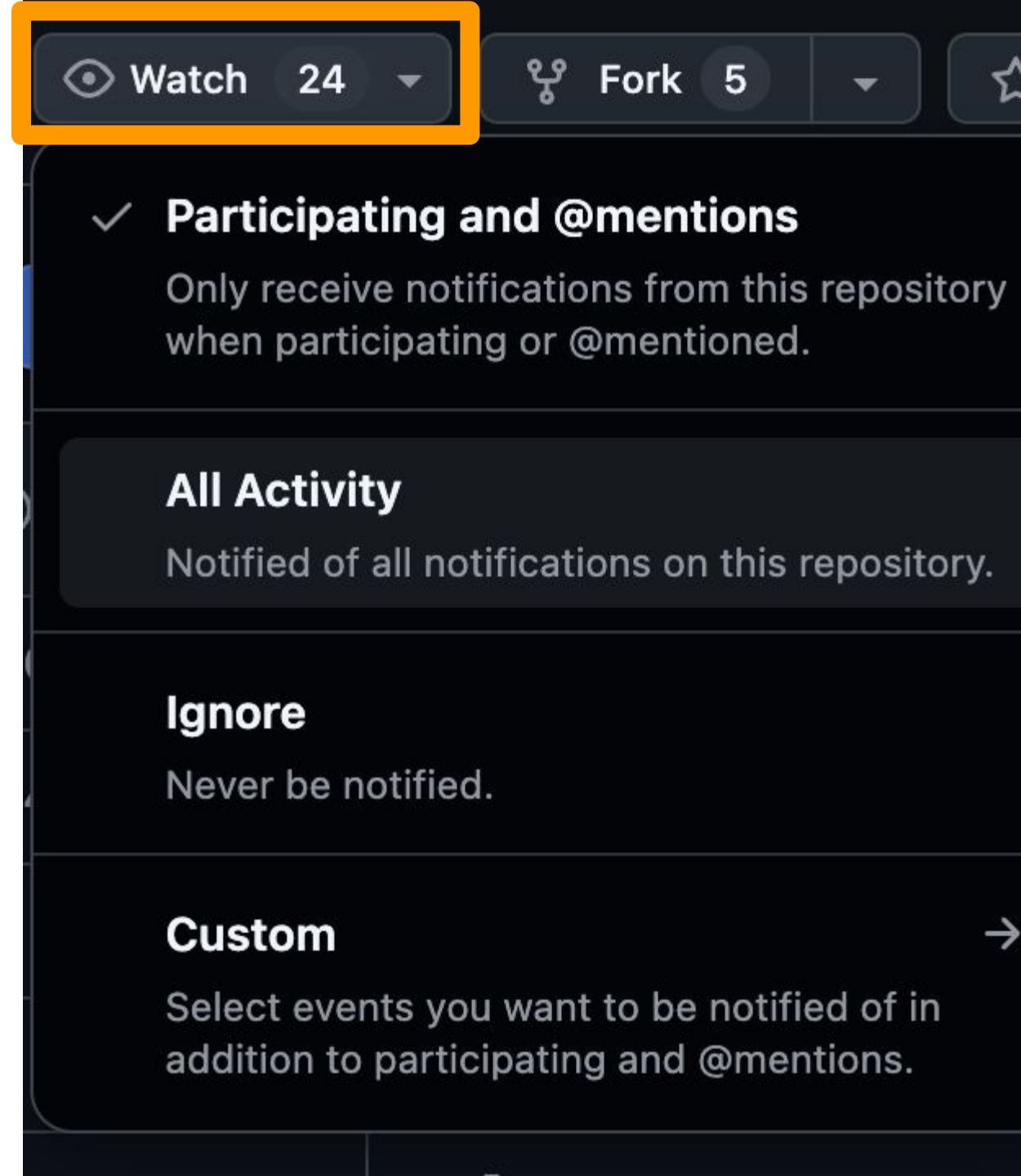
Next steps

How to be involved?

1) Keep updated:

- watch the Camtrap DP [GitHub repo](https://github.com/tdwg/camtrap-dp)¹ to get notifications

¹ <https://github.com/tdwg/camtrap-dp/issues>



How to be involved?

1) Keep **updated**:

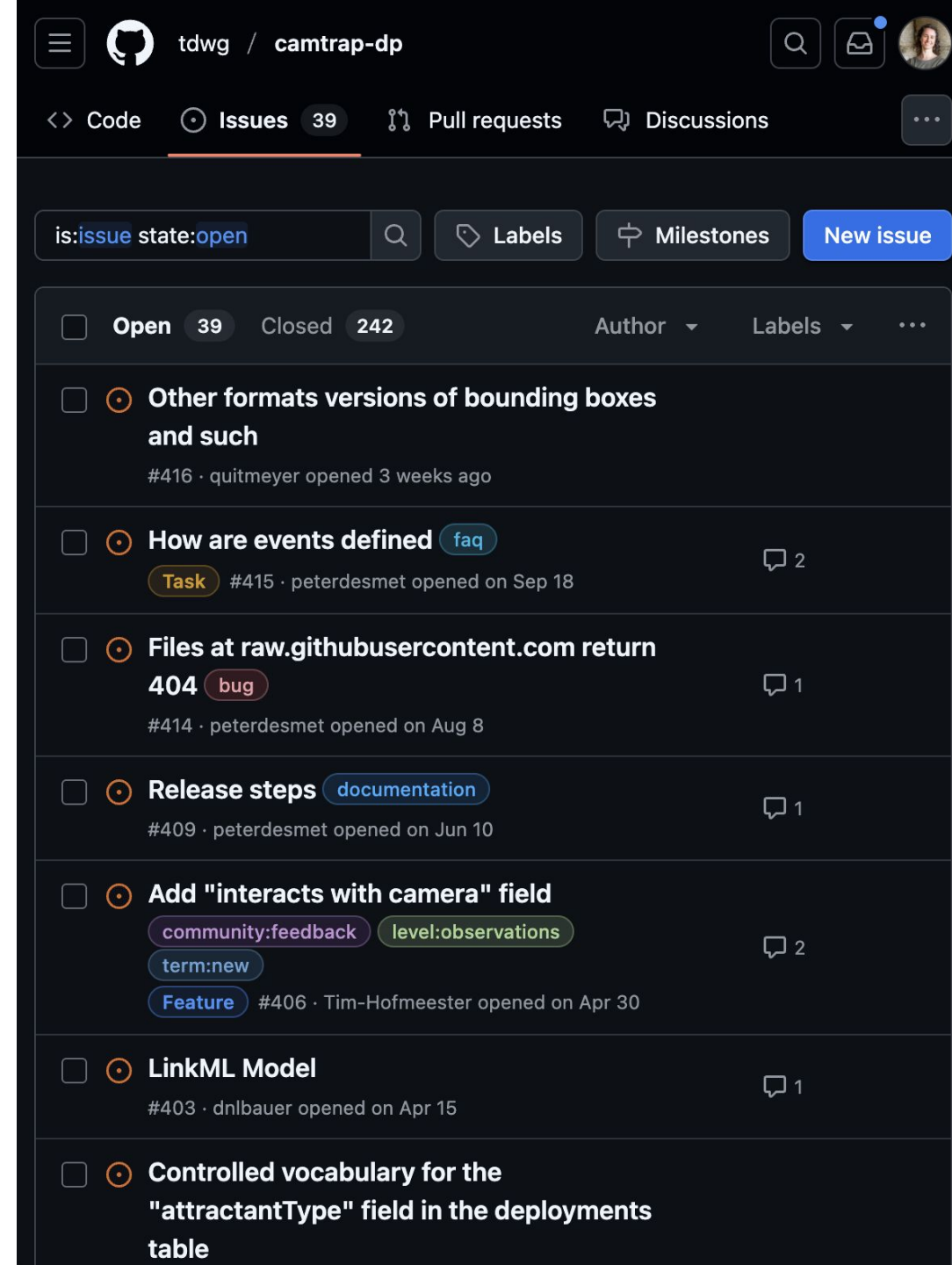
- watch the [GitHub repo](https://github.com/tdwg/camtrap-dp)¹

2) **Contribute** to:

- the discussion in [GitHub issues](https://github.com/tdwg/camtrap-dp/issues)¹

If you have difficulty using Github comment system, you can email comments to sanne.govaert@inbo.be (EN, NL) or emarentes@humboldt.org.co (ES, EN) and we will post them on your behalf

¹ <https://github.com/tdwg/camtrap-dp/issues>



How to be involved?

- 1) **Keep updated:**
 - watch the [GitHub repo](https://github.com/tdwg/camtrap-dp/issues)¹
- 2) **Contribute to:**
 - the discussion in [GitHub issues](https://github.com/camera-traps/bioacoustics)¹
- 3) **Test your data:**
 - [bioacoustic data](https://github.com/camera-traps/bioacoustics)²
 - [insect data](https://github.com/camera-traps/insectai)³

¹ <https://github.com/tdwg/camtrap-dp/issues>

² <https://github.com/camera-traps/bioacoustics>

³ <https://github.com/camera-traps/insectai>



Photo by Julia Wiel

Thank you!

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- Norwegian Institute for Nature Research
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Engystomops pustulosus, photo by Nicolás Betancourt

References

- Bubnicki, Jakub W., Ben Norton, Steven J. Baskauf, Tom Bruce, Francesca Cagnacci, Jim Casaer, Marcin Churski, et al. 'Camtrap DP: An Open Standard for the FAIR Exchange and Archiving of Camera Trap Data'. *Remote Sensing in Ecology and Conservation* 10, no. 3 (June 2024): 283–95. <https://doi.org/10.1002/rse2.374>.